

Development task breakdown

EPIC 0 – PoC Governance & Alignment

Goal: Avoid scope creep and misalignment.

- Define PoC success metrics (latency, cost, parity)
 - Confirm GLM coefficients format from SPSS with the Client
 - Confirm LTV/CMA dataset availability & KPIs with the Client
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EPIC 1 – AWS & SageMaker Foundation

Story 1.1: Storage & security

- Create S3 buckets:
 - `raw/`
 - `processed/`
 - `features/`
 - `models/`
 - `batch-input/`
 - `batch-output/`
 - Enable encryption, access logging, tagging
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EPIC 2 – Data Ingestion (IBM SPSS → AWS)

Story 2.1: Data contracts

- Define schemas for GLM, LTV, CMA datasets
- Define versioning strategy

Story 2.2: Data ingestion pipeline

- Export datasets from IBM SPSS / Google DB
 - Upload to S3 (versioned) - Need to write a script (What is the size and duration for this copy activity ?)
 - Data integrity checks
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EPIC 3 – Data Preprocessing (SPSS parity)

Story 3.1: Preprocessing logic

- Missing value handling
- Outlier treatment
- Encoding strategies
- Normalization

Story 3.2: Validation

- Compare SPSS vs SageMaker outputs - What is the sampling size ?
 - Document differences (if any)
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EPIC 4 – Feature Engineering Layer (Shared between GLM, LTV, CMA)

Story 4.1: Feature modules

- Centralized feature code
- Reusable for training & inference

- Version-controlled

Story 4.2: Feature consistency tests

- Unit tests for training vs serving
 - Schema validation
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EPIC 5 – GLM Inference (Rule-Based)

Story 5.1: GLM migration

- Import coefficients from SPSS
- Implement frequency & severity logic
- Compute Base Cost = $Fr \times Sev$

Story 5.2: Explainability

- Log intermediate outputs
 - Validate actuarial expectations
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EPIC 6 – LTV Model (Training + Inference)

Story 6.1: Training implementation

- Define LTV objective (KPI alignment)
- Implement training script
- Configure SageMaker Training Job

Story 6.2: Manual training

- Trigger job via Notebook / Console
- Capture metrics

Story 6.3: Model registration

- Register in Model Registry
 - Store metrics, data version
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EPIC 7 – CMA Model (Training + Inference)

- Implement CMA training script
 - Run manual training
 - Register model
 - Validate integration into pricing
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EPIC 8 – Model Registry & Promotion

Story 8.1: Promotion workflow

- Model versioning
 - Rollback strategy
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EPIC 9 – Pricing Engine (Core Logic)

Story 9.1: Tariff calculation

- Combine:
 - GLM base cost
 - LTV adjustment
 - CMA adjustment
 - Business rules

Story 9.2: Rules abstraction

- Config-driven rules
 - No hardcoding
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EPIC 10 – Real-Time (1:1) Endpoint

Story 10.1: Endpoint deployment

- SageMaker real-time endpoint
- REST interface
- Stateless inference

Story 10.2: Performance testing

- p50 / p95 latency
 - Concurrent load tests
 - Autoscaling tuning
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EPIC 11 – Batch Pricing Pipeline

Story 11.1: Batch Transform setup

- Batch input/output via S3
- Parallel execution

Story 11.2: Cost & performance validation

- Runtime benchmarking
 - Cost per batch
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EPIC 12 – Monitoring & Observability

Story 12.1: Metrics & logs

- Latency
- Errors
- Throughput

Story 12.2: Alerts

- Endpoint failures
 - Batch job failures
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EPIC 13 – Security Hardening

- API security
 - Access audit logs
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EPIC 15 – PoC Validation & Demo

Story 15.1: Benchmarking

- SPSS vs SageMaker parity
- Latency & cost comparison

Story 15.2: Demo scenarios

- GLM coefficient update
- LTV retrain
- CMA retrain
- Real-time + batch together

Story 15.3: Documentation

- Architecture
- Findings
- Trade-offs